

Hardening Transcriptomic Prognostic Evidence in Clear Cell Renal Cell Carcinoma

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Abstract

Which tumor-normal transcriptomic associations in clear cell renal cell carcinoma (ccRCC) retain prognostic support under stronger validation? We used TCGA-KIRC for discovery and GSE40435 and GSE53757 for expression replication. Previously inspected GSE29609 and E-MTAB-1980 supplied external survival assessments. Expression replication, clinically adjusted survival models, and stricter sensitivity and effect gates reduced 8,534 discovery genes to 23 priority associations. These 23 retained conditional bootstrap support, but adding a gene selected within training folds changed held-out concordance by only 0.0047 in 517 patients, with a conditional 95% interval of -0.0079 to 0.0174. Survival direction agreed for 5/21 mapped candidates in GSE29609 versus 21/22 in E-MTAB-1980; two associations reversed with false-discovery-rate support in GSE29609. Marker-score adjustment removed support for 6 associations, whereas consensus-purity adjustment retained all 23. The workflow exposes disagreement between expression reproducibility, prognostic association, and prediction. It supplies auditable evidence for follow-up, without establishing a transportable prognostic panel or clinical utility.

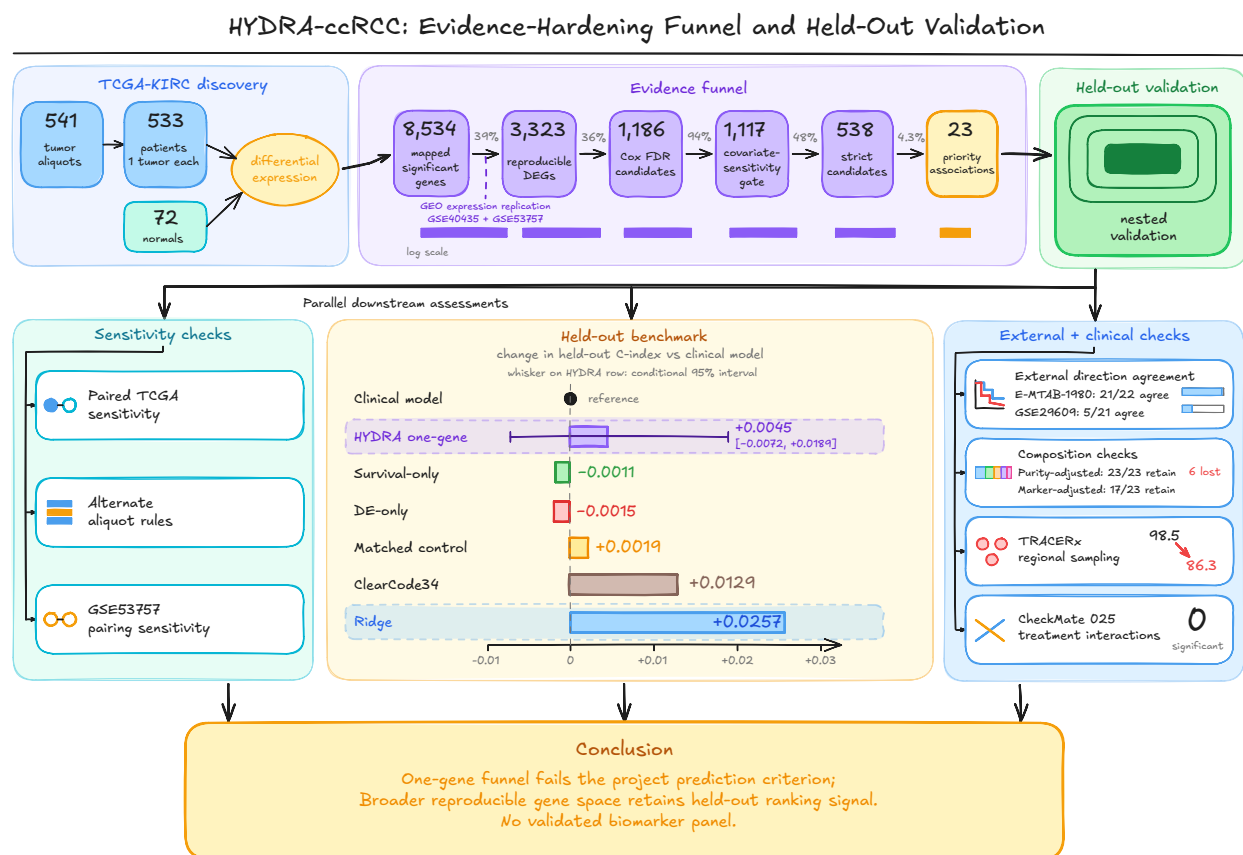


Figure 1: HYDRA evidence overview. The preserved author artwork shows the six selection gates and parallel downstream assessments. Its prediction labels precede the normalization correction; current estimates appear in the results and supplement. External direction counts use unadjusted models. The HYDRA whisker is a conditional patient-bootstrap interval from saved benchmark predictions. Expression replication and prognostic support do not establish clinical utility.

1 Introduction

A gene can distinguish a ccRCC tumor from normal kidney without improving prognosis beyond clinical stage and grade. HYDRA asks one primary question: *which tumor-normal transcriptomic associations retain prognostic support as replication, clinical adjustment, held-out evaluation, and tissue-composition checks strengthen the evidence?* Three secondary questions ask whether survival associations agree across cohorts, whether the selection procedure improves prediction beyond clinical variables, and whether composition changes the interpretation of the associations.

The contribution is a reproducible workflow and a multi-cohort evidence audit. Familiar methods supply different tests of the same candidate claims, and a gene-level ledger preserves disagreements instead of turning a discovery shortlist into a validated panel. Differential expression, prognostic association, prediction performance, and biological plausibility are reported separately. Clinical utility is not evaluated. HYDRA introduces no statistical algorithm.

1.1 Relation to biomarker-discovery practice

REMARK asks investigators to report patient selection, covariates, assay methods, and validation clearly [9]. Selection-aware validation is also established: Simon and colleagues require feature selection to occur inside resampling when estimating a discovery procedure’s performance [12]. HYDRA applies these principles rather than proposing a new estimator. Its additional output is a linked record of how the same claims change under stronger evidence, including negative cohort results and competing composition explanations.

ClearCode34 developed a ccRCC subtype classifier and evaluated it in TCGA and independent clinical specimens [3]. HYDRA instead audits a tumor-normal-first prognostic procedure. Its signed ClearCode34 comparator is an adaptation for these splits, not a reproduction of the published classifier or a clinical head-to-head evaluation. The contribution beyond running familiar analyses sequentially is the explicit separation of selection gates from downstream assessments, a complete gene ledger, and comparisons of research procedures on common splits and matched list sizes. Whether this workflow produces better biomarkers remains an empirical question; the present comparisons do not establish general superiority.

2 Evidence design

TCGA-KIRC supplies 533 primary-tumor patients and 72 normal-kidney samples. One aliquot per patient is selected by library depth. The primary survival analysis includes 517 patients and 170 deaths after complete-case exclusions. GSE40435 and GSE53757 supply 101 and 72 tumor-normal pairs, respectively. GSE29609 contains 39 patients and 17 deaths; E-MTAB-1980 contains 101 patients and 23 deaths. These two survival cohorts were inspected before the current analysis and are exploratory checks, not untouched validation. Dataset identities and complete inclusion rules appear in the supplement and cohort dictionary.

The evidence funnel has six selection gates. TCGA differential expression requires false discovery rate (FDR) below 0.05 and absolute $\log_2$ fold change at least one. Expression replication requires the same direction in both GEO cohorts and nominal support in at least one. Clinically adjusted Cox models then test continuous expression alongside age, sex, stage, and grade. Covariate sensitivities remove stage or grade separately. Two increasingly strict gates add effect-size and FDR thresholds. The final 23-gene set is a priority shortlist of associations, not a validated panel. Subsequent analyses assess this set without redefining it.

We distinguish four kinds of evidence. Cox coefficients describe prognostic association. Repeated held-out comparisons assess prediction performance beyond a clinical model. Purity and normal-tissue cell-source data inform biological interpretation. None evaluates clinical utility, which would require a defined clinical decision and prospective evidence. Treatment-effect modification is tested separately in the supplement.

For prediction, ten repeats of five patient folds rerun TCGA selection and training-only preprocessing. The GEO expression evidence remains fixed. Each fold adds its top qualifying gene to a clinical model. The project criterion requires mean concordance improvement of at least 0.01, a conditional patient-bootstrap interval above zero, and no worse three-year Brier score. The bootstrap resamples saved predictions and does not refit training models or redraw splits. Full-data coefficient bootstraps also condition on the selected shortlist. Model specifications, testing families, seeds, and secondary checks are in the supplement.

3 Results along the evidence funnel

3.1 Discovery contracts to a small association shortlist

The funnel contracts from 8,534 mapped TCGA discovery genes to 3,323 expression-reproducible genes, 1,186 adjusted survival associations, 1,117 passing covariate sensitivity, 538 strict candidates, and 23 priority associations. Figure 1 shows the six selection gates and parallel downstream assessments; the counts are audited in the ledger. The ledger records each mapped gene, including failed gates and explicit missing downstream evidence, so favorable candidates cannot conceal the denominator.

Changing aliquot rules leaves 19, 20, or 19 priority genes instead of 23. This sensitivity shows that the shortlist boundary depends on sample handling. The unpaired GSE53757 analysis retains all 23 genes, but agreement between fits cannot verify the assumed patient pairs. Full gene tables and both sensitivities are reported in the supplement.

3.2 Survival evidence depends on the cohort

E-MTAB-1980 retains the TCGA hazard direction for 21 of 22 mapped genes; 12 meet the directional FDR rule in both unadjusted and limited-adjustment models. Unadjusted GSE29609 models retain the direction for only 5 of 21, with no same-direction nominal support. DDC and TCIRG1 have opposite-direction FDR support there. Cohort size, platform, and clinical adjustment differ, so these results describe disagreement rather than identifying its cause.

Harmonized age-and-grade adjustment leaves 6 supported E-MTAB associations; adding T category leaves 1. The two GSE29609 reversals retain opposite directions but lose FDR support after adjustment. Thus, comparable adjustment weakens external support rather than resolving cohort disagreement.

ACADM illustrates agreement between TCGA and E-MTAB-1980, while its GSE29609 estimate points in the opposite direction with uncertainty spanning no association. Its E-MTAB association loses FDR support after age, grade, and T adjustment. Figure 2 shows every cohort rather than presenting the favorable comparison alone. The primary conclusion is therefore cohort-specific support, not uniform replication.

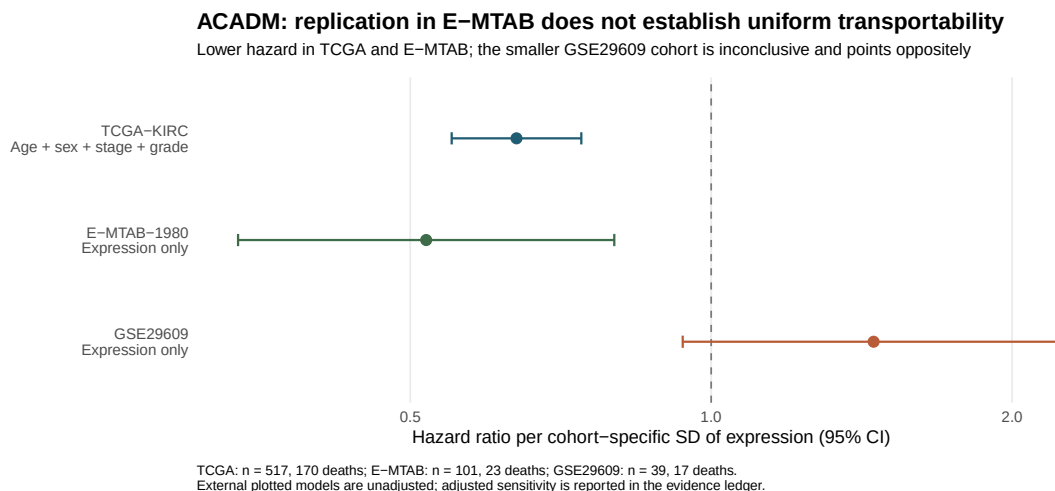


Figure 2: Representative cohort-specific agreement for ACADM. Hazard ratios and 95% intervals use continuous expression. TCGA adjusts for age, sex, stage, and grade; external estimates are unadjusted. Standardization uses cohort-specific expression scales, so effect magnitudes are not a common assay-scale contrast. E-MTAB-1980 supports the TCGA direction, while GSE29609 does not. Both external cohorts were previously inspected.

3.3 Conditional association stability does not establish prediction

All 23 full-data selected genes have coefficient-bootstrap intervals excluding zero. That result addresses uncertainty after selection; it does not estimate stability of the selection procedure. Within the 50 training folds, the top gene varies across 9 genes, and the most frequent appears in 22 folds. Adding the selected gene changes concordance by 0.0047 with conditional 95% interval -0.0079 to 0.0174, and three-year Brier score by -0.0010. The procedure fails its project criterion: the mean concordance gain is below 0.01 and its interval spans zero. This does not establish that the underlying associations are null.

Existing procedure comparisons quantify how discovery choices change the evidence. At an equal requested list size of 22, DE-only selection versus the complete rule gives direction agreement of 18/20 versus 21/21 mapped genes in E-MTAB-1980, but 5/16 versus 4/20 in GSE29609. Same-direction FDR support is 3 versus 13 in E-MTAB-1980 and zero for both in GSE29609. The ranking rules and mapping denominators differ, and outcomes were previously inspected. These comparisons do not show that stricter filtering reliably improves transportability.

On identical nested TCGA splits, DE-only, survival-only, and HYDRA one-gene procedures change concordance by -0.0014, -0.0010, and 0.0047, respectively. Across those 50 folds, DE-only selection chooses 5 distinct genes versus 9 for HYDRA, so stronger filtering does not make gene identity invariant. A separate ridge model using all training-fold reproducible genes changes concordance by 0.0257. Its Brier difference interval spans zero, and it has not been evaluated externally. This exploratory multigene comparison shows held-out ranking information within TCGA without rescuing the failed one-gene criterion or establishing a signature. Full uncertainty and comparator specifications are in the supplement.

3.4 Composition changes the interpretation of some associations

Consensus-purity adjustment on matched patients retains all 23 hazard directions and FDR associations. Bulk marker-score adjustment removes FDR support for 6, leaving 17. DDC illustrates the tension: its GSE29609 association reverses, and marker adjustment attenuates its TCGA coefficient, whereas purity adjustment barely changes it. Figure 3 reports both results rather than choosing one composition measure as definitive.

4 Discussion and limitations

The strongest result is the mismatch between a reproducible association shortlist and the evidence needed for a transportable prognostic claim. Expression replication and covariate adjustment yield 23 associations, yet cross-cohort directions disagree, the selected one-gene procedure adds little held-out discrimination, and some associations depend on the composition adjustment. The workflow makes these differences inspectable. It does not prove that evidence hardening is universally superior to simpler discovery procedures.

Post hoc sensitivities address missing covariates, collection site, technical plates, and unequal cohort adjustment without redefining the shortlist. All 23 associations retain direction and FDR support across eight explicit covariate-completion scenarios, and 23 retain support after stratification by collection site. Plate-adjusted tumor-normal models retain the primary DE gate for 19 candidates across all plates and 19 on the three plates with both tissue types. These losses reflect the fold-change cutoff; direction and FDR support remain for all 23.

Several threats remain. Complete-case analysis excludes 16 TCGA patients; completion scenarios do not model missingness or restore zero-time outcomes. GSE53757 pair identities are inferred from adjacent sample order. Plate labels are incomplete technical proxies, so unwanted variation remains possible. The accession screen detects no shared identifiers, but cannot establish patient independence. Different platforms, clinical contexts, and small external event counts still limit transportability.

Benjamini-Hochberg (BH) correction applies within defined testing families, not to every adaptive choice across the workflow. The expression screen and repeated project development limit interpretation of nominal FDR guarantees. 2 primary gene models have nominal proportional-hazards flags; non-significant shape diagnostics cannot establish proportional hazards. Corrected held-out normalization uses a frozen training reference without test-batch re-centering. Conditional bootstrap intervals still omit selection and training

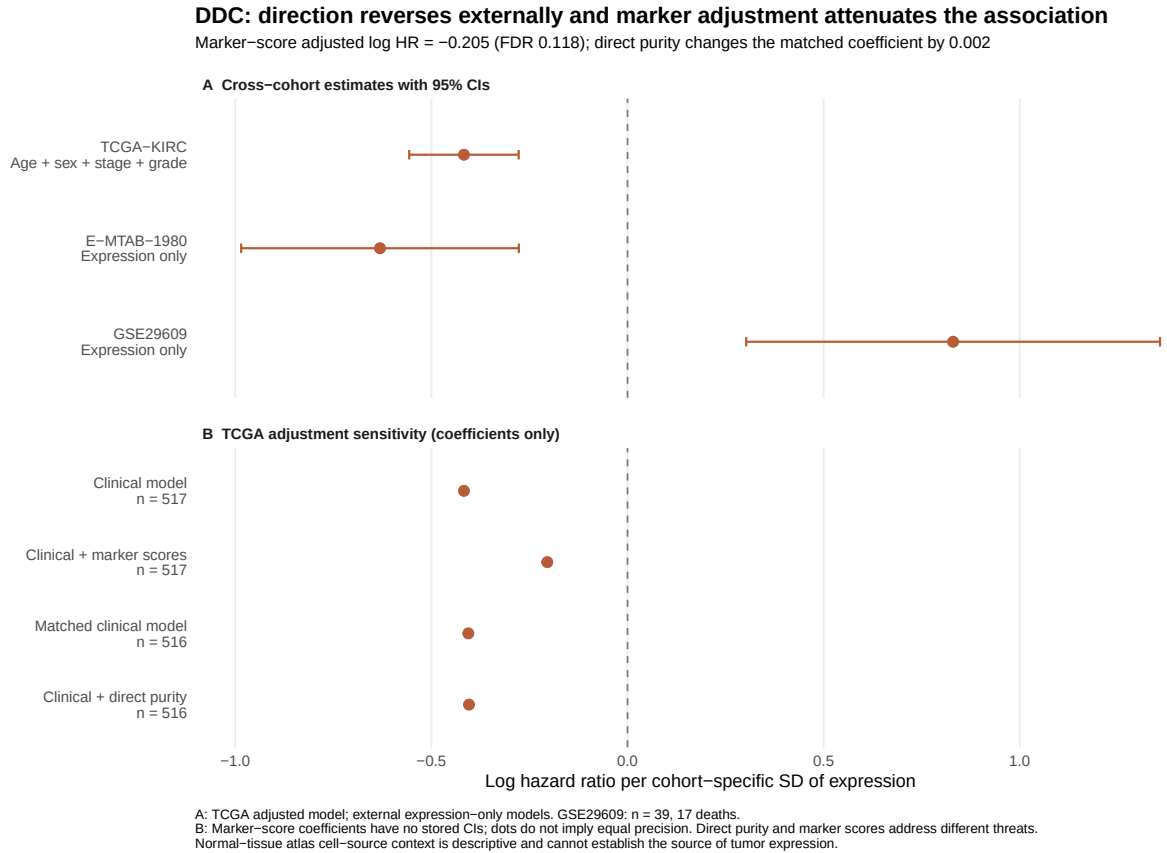


Figure 3: DDC illustrates external reversal and composition sensitivity. Cohort hazard ratios carry saved 95% intervals. Matched-patient composition comparisons show saved coefficients and FDRs; marker-adjustment confidence intervals were not stored and are not inferred. Marker scores are expression proxies, not measured cell proportions. Attenuation does not identify causal mediation, and retained purity-adjusted support does not establish tumor-cell origin.

uncertainty. Human Protein Atlas (HPA) describes normal-tissue cell sources, not ccRCC tumor cells. Previously inspected outcomes and bulk composition proxies cannot establish causality, prospective validation, or clinical utility. No decision threshold, prospective assay, or patient-benefit evaluation is available.

Data availability and reproducibility

Public inputs are identified by accession and source URL, with a pinned TRACERx commit, cached-input hashes, package versions, and seeds. Original retrieval dates were not recorded and remain unknown. The revision records current execution dates separately from original data access. The candidate ledger and claim checks are generated by scripted commands; the source-to-output manifest identifies regenerated tables and figures. Cached data are excluded from Git and can be retrieved through the documented pipeline. Full procedures, gene tables, and secondary checks follow in the supplement.

Ethics

This secondary analysis uses public de-identified datasets and involves no new recruitment or intervention. No new institutional ethics approval is claimed; applicability of local requirements requires author review.

Author contributions

Arjun Vijay Prakash: conceptualization, data curation, software, formal analysis, visualization, and writing.

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AI use

OpenAI Codex assisted with code, artifact auditing, figures, and manuscript revision. AI assistance is part of manuscript preparation, not the scientific object of study. The author is responsible for the analyses, citations, interpretation, and final text.

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A Supplementary methods

A.1 Inputs, patient identity, and discovery

TCGA-KIRC STAR unstranded counts and clinical outcomes were retrieved through TCGAbiolinks [4]. For each patient and sample type, we selected the aliquot with greatest raw-count library depth and broke barcode ties so the choice is deterministic rather than biological. The 541 tumor aliquots came from 533 patients, including 175 deaths: four patients contributed three aliquots each, which accounts for all eight extra aliquots. Of those 533 patients, 16 are absent from the primary Cox model and the nested splits because each fails exactly one required field—8 lack grade, 3 lack stage, 1 lacks age, and 4 have overall-survival time recorded as zero—leaving 517 patients and 170 deaths (5 of the excluded patients died). The stage-only sensitivity restores the grade-missing patients (525) and the grade-only sensitivity restores the stage-missing patients (520). All 72 normals were already unique by patient. Genes are matched by symbol after a dataset-specific collapse, recorded in the gene-mapping table. TCGA strips the Ensembl version, maps the id with org.Hs.eg.db, and keeps the first symbol. GSE40435, GSE53757, and GSE29609 take the first token of the platform gene-symbol field and average probes that share it. E-MTAB-1980 maps RefSeq identifiers the same way and averages rows that share a symbol. TRACERx matches candidate symbols to expression rownames. CheckMate 025 averages rows that share `gene_name`. Survival-event coding was checked against vital status and the appropriate death or follow-up date; joins were required to remain one row per patient. Three alternate gates, reported as sensitivities, dropped those four patients, kept the first barcode, or summed a patient’s tumor counts. Original retrieval dates for cached public inputs were unavailable, so the repository records source URLs and hashes without assigning new access dates.

DESeq2 compared the selected tumors and normals after requiring at least ten counts in ten samples [8]. A separate patient-blocked DESeq2 fit, with design patient plus tissue type, used the 72 selected tumor-normal pairs. GSE40435 and GSE53757 were analyzed with patient-blocked limma; each parsed patient was required to have exactly one tumor and one normal, and surrogate-variable analysis assessed unwanted variation while protecting the paired contrast [7, 10]. GSE53757 has no field that stores one shared patient id for every pair. Adjacent rows were opposite tissues and the same stage in all 72 pairs, and 4 pairs also shared a title token, so the primary pair ids follow that adjacent order. An unpaired limma fit is a sensitivity. A reproducible DEG required TCGA FDR below 0.05 and absolute $\log_2$ fold change at least one, the same sign in both GEO cohorts, and nominal support in at least one, defined as a two-sided unadjusted $p < 0.05$ with that same sign. An all-gene Cox sensitivity removed the upstream fold-change gate and retained apeglm-shrunken differential-expression estimates as annotations. It was not allowed to redefine the primary candidate set [13].

The primary Cox model standardized expression over the eligible tumor survival cohort before complete-case covariate exclusion and used age, sex, categorical AJCC stage with Stage I as the reference, and grade collapsed to G1+G2 versus G3+G4 among complete cases [5]. G1 contains 14 patients in the TCGA clinical table, so it was not kept as its own level. The strict gate also required main-model FDR below 0.05, absolute log hazard ratio at least $\log(1.25)$, both GEO effects at least 0.25 in absolute $\log_2$ units, and concordant nominal support in two sensitivity fits: expression plus age, sex, and stage, and expression plus age, sex, and grade. The high-confidence gate required FDR below 0.01 and absolute log hazard ratio at least $\log(1.5)$. Those magnitude cutoffs are fixed project thresholds, not a cited minimal clinically important difference. False-discovery correction stays inside one family at a time: differential expression, each GEO cohort, Cox models among reproducible genes, and the high-confidence tests for shape, composition, and treatment interaction. Proportional-hazards tests were diagnostics, never selection gates. The selected gene is the one with the highest evidence score, $-\log_{10}(\text{FDR}) + |\log \text{HR}|$ plus bounded fold-change terms, with remaining ties broken by FDR and then gene id. The acceptance test adds that one gene. The ridge fit is a separate multigene description. It does not replace the failed one-gene criterion or freeze a signature. It is a glmnet Cox model with Efron ties, clinical covariates unpenalized, and an L2 penalty ($\alpha = 0$) on every training-fold reproducible gene with nonzero training standard deviation. The penalty does not zero coefficients and does not name a gene subset. Before any penalty is chosen, the outer training fold fixes the design matrix: DESeq2 size factors are estimated on the training samples, test size factors use that training geometric-mean reference, expression is $\log_2(\text{count}/\text{size factor} + 1)$, and genes with zero training standard deviation are dropped. Clinical factor levels come from the 517-patient complete-case cohort; the values in

a fit come from its training rows. λ is then chosen by an inner cross-validation that uses only those training patients. `cv.glmnet` first fits the λ path on all outer-training rows, scores mean partial-likelihood deviance on five event-stratified inner folds of those same rows, and keeps $\lambda_{\min}$. The criterion is deviance, not the concordance index and not the one-standard-error rule. Each inner fit standardizes its own training columns. The coefficients applied to the outer test rows come from the refit on the full outer training fold, and that fold’s column means and standard deviations are applied to the test rows. Outer test outcomes are not an input. The inner-fold seed is the project seed plus $31000 + 10 \times \text{repeat} + \text{fold}$, drawn before `cv.glmnet`. The reported contrast compares this joint fit with a separate clinical Cox model. Clinical coefficients inside the ridge fit are estimated together with the penalized genes, so the concordance increment is not an isolated gene effect and does not identify which genes produce the gain.

A.2 Selection-aware prediction and null simulation

Ten repeats of five event-stratified patient folds reran TCGA DE and every candidate gate within each training fold of the 517 complete-case patients defined above. Fold labels were redrawn for each repeat. Event stratification shuffled deaths and censored patients separately, and every outer test fold contained 34 deaths. The full-data stage-only and grade-only sensitivities contain 525 and 520 patients, respectively, and are therefore not numerically identical to these nested fits. GEO expression evidence remained fixed; test patients and their normals were excluded from training. That is the scope of “selection-aware” here: TCGA differential expression, the fixed GEO gate, the Cox gates, and the choice of one gene. It does not re-split the GEO cohorts or re-choose the 0.01 rule inside a fold. Library-size normalization and expression scaling were estimated on training samples and applied to held-out tumors. The nested fit used $\log_2$ normalized counts, while the full-data survival analysis used a variance-stabilizing transformation; this preserves training-only preprocessing but changes the expression scale, limiting direct coefficient comparisons. DESeq2 retained the primary outlier-replacement setting, with a logged no-replacement retry only if the pinned runtime failed in that step. This fallback can change gene selection. Comparator regeneration replays the replacement setting recorded for each primary fold and retries that setting rather than switching it independently. The top qualifying gene was added to a clinical Cox model; when no gene qualified, its prediction equaled the clinical-only prediction. Out-of-fold concordance, three-year inverse-probability-of-censoring-weighted Brier score, and risk-score calibration slope were computed per repeat. The concordance increment received a patient-resampled 95% interval conditional on the saved out-of-fold predictions; it does not include variation from refitting selection or redrawing folds. The project success rule required an increment at least 0.01, a conditional interval above zero, and a mean Brier difference no greater than zero. The 0.01 increment is this project’s size requirement, not a universal clinical standard. A lower Brier score is not, by itself, better calibration: discrimination and calibration contribute separately to the Brier score.

Two hundred clinical-only null datasets used an exponential baseline calibrated to the clinical Cox cumulative hazard with resampled censoring; each reran the full training selection on one held-out fold, cycling across the five fixed folds from repeat 1. This is a parametric clinical-risk null, not another ten-by-five-fold interval and not a p-value for the concordance increment. A second null permuted training survival pairs on the five saved repeat-1 differential-expression fits, without refitting DESeq2, and cycled the same way across 200 simulations. The original 1,000 patient bootstraps and 20-repeat per-gene CV condition on the full-data shortlist and remain descriptive rather than selection-adjusted evidence.

The same patients, seeds, and event-stratified folds were then used to compare selectors. One training-only DESeq2 fit supplied six predictors for each held-out fold: the clinical model; the primary HYDRA gene; a survival-only gene held to the same high-confidence hazard gates without tumor-normal or GEO filters; a differential-expression-only gene held to the training DESeq2 thresholds without survival or GEO filters; a ridge Cox model on every training-fold reproducible DEG, with clinical covariates unpenalized and $\lambda_{\min}$ chosen by the training-only inner deviance cross-validation above [11]; and one expression-matched gene at or below the median residual correlation with the clinical martingale residuals. Survival-only false-discovery rates were computed as if every unscreened count-QC gene had $p = 1$, after fitting the top 2,000 residual correlations. The HYDRA gene was required to match the saved primary nested result. Intervals remain conditional on the saved predictions. This comparison does not replace the project one-gene acceptance rule.

On those same folds we also added a signed-expression adaptation of the published 34-gene ClearCode34 classifier [3]. ccA genes were signed negative and ccB genes positive. Current symbols were taken from gene

aliases where the published symbol has changed; one published symbol did not map. Expression was $\log_2$ counts per million, and the training fold supplied the mean and standard deviation. This arm does not select genes. Separately, on the previously inspected E-MTAB-1980 cohort, we fit in-sample Cox models for the study’s ccA/ccB label, for the highest-evidence primary gene present on that array (CRYL1; CYFIP2 did not map), and for a sign-weighted mean of the mapped high-confidence genes, each added to the limited clinical model. Those E-MTAB fits are not nested, and their in-sample calibration slopes are not held-out estimates.

A.3 Cohort-specific survival models

The post hoc limitation audit keeps the 23-gene shortlist fixed and adds the same three models in each cohort: expression alone, expression plus age and collapsed grade, and expression plus age, grade, and T category. Grade is G1–G2 versus G3–G4; T is T1–T2 versus T3–T4. Sex is omitted from the harmonized fits because it is unavailable in the supplied GSE29609 clinical fields. Expression is standardized separately within each cohort before model-specific exclusions. Each cohort/model uses a 23-hypothesis BH family, including unavailable mappings. GSE29609 has complete age, grade, and T data; two E-MTAB patients lack numeric grade, and the adjusted analyses use 99 patients with 23 deaths. TCGA has 529 positive-time patients before covariate exclusion. The harmonized fits include confidence intervals, patient/event counts, PH diagnostics, and fitting warnings. They remain exploratory comparisons of previously inspected cohorts. Q and I2 describe variation in cohort-standardized coefficients, not a pooled biological effect; scaling and platforms also contribute to heterogeneity. The Q calculation assumes independent cohort estimates, which the accession screen cannot establish.

Eight explicit TCGA completion scenarios replace missing grade with low/high grade, missing stage with I/IV, and missing age with the cohort’s 10th/90th percentile. They retain positive-time patients and refit the primary adjustment on the fixed shortlist. Completion and collection-site fits each use the fixed 23-gene BH family. These scenarios are not multiple imputation, mathematical worst-case bounds, or a model of the missingness mechanism. A collection-site sensitivity stratifies the 517 complete cases by barcode-derived source site. Sites with fewer than ten patients are pooled into one small-site stratum, producing 11 strata. This allows different baseline hazards by collection site but does not identify or remove technical batch effects. A literal accession-ID screen across the five expression cohorts found no shared IDs; different identifiers for the same patient remain possible.

The technical-plate sensitivity uses the four-character plate field of the selected TCGA aliquot barcodes. DESeq2 fits plate plus tissue type on all 605 samples and separately on the three mixed-tissue plates, containing 145 tumors and 68 normals. Both designs must have full column rank. The original count filter, global transcript-level BH adjustment with DESeq2 independent filtering, and DE magnitude threshold remain in force. The results annotate the fixed shortlist rather than select new candidates. The all-plate contrast draws its within-plate tissue comparison from the mixed plates; the restricted fit also changes the sampled population. Plate labels do not capture every processing batch. Both fits report outlier-replacement fallback status and all tested transcripts.

The normalization audit reproduced all 50 original primary predictions to numerical tolerance before changing held-out transforms. DESeq2’s supplied-reference API re-centers size factors within the supplied batch. The correction uses the same median ratios against the training geometric means without this test-batch re-centering, making an individual sample’s transform invariant to its companions. Training expression and gene selections were unchanged, and Cox prediction models were refitted. Original outputs and checkpoints are preserved under `results/archive/normalization_20261004`; the machine-readable audit records the original hashes and prediction differences. Benchmark models are refitted because their fitted objects were not retained. This correction does not make the saved-prediction bootstrap unconditional. The primary external estimates are unadjusted continuous-expression Cox models. GSE29609 also fits age, high grade, and high T-stage, whereas E-MTAB-1980 fits age, high T-stage, and metastatic status. These covariate sets differ from TCGA’s age, sex, categorical stage, and collapsed-grade adjustment. The strict E-MTAB support flag requires the TCGA direction and FDR below 0.05 in both external models; it does not exclude genes by proportional-hazards diagnostics. GSE29609 did not save adjusted confidence intervals, adjusted sample/event counts, or PH tests, so those diagnostics are unavailable in the ledger. These models are exploratory because both cohorts were previously inspected.

A.4 Exploratory alternatives and biological checks

At the largest common list size of 22 genes, we compared the complete rule with survival-only, DE-only, each leave-one-GEO-out rule, and TCGA-expression-matched non-prognostic controls. GSE29609 and E-MTAB-1980 were previously inspected, so both evaluations are exploratory. We estimated the complete-rule minus survival-only directional-rate difference and a 2,000-draw paired gene bootstrap in each cohort. The project funnel criterion required an interval entirely above zero.

Candidate Cox models were refit on matched complete cases with published TCGA consensus tumor purity [1] or leave-one-candidate-out marker scores. Each score is the mean of sample-standardized marker expression, then standardized across samples. The sets are proximal tubule (AQP1, LRP2, CUBN, SLC5A2, ALDOB), endothelial (PECAM1, VWF, KDR, EMCN, TEK), immune (PTPRC, CD3D, CD8A, MS4A1, CD68), and stromal (COL1A1, COL1A2, DCN, LUM, ACTA2). A candidate that belongs to a set is omitted from that score. These are bulk marker scores, not measured cell proportions. Consensus purity is an inferred bulk estimate. Nonlinear expression and expression-by-log-time models were tested against the linear proportional-hazards fit. Human Protein Atlas normal-tissue single-cell profiles informed cell-source cautions but were not treated as ccRCC tumor single-cell validation.

TRACERx Renal provided primary-region expression and linked survival data from a pinned public repository [6]. One thousand repeats selected one region per patient. A fixed event-stratified 39-patient subset isolated region-choice variation; a separate scenario resampled which 39 patients entered each repeat. Check-Mate 025 supplementary RNA data supported post hoc continuous-expression-by-treatment Cox interactions for overall and progression-free survival, with FDR correction within endpoint and model [2].

B Supplementary results

B.1 Corrected discovery and conditional sensitivity

533 patients to 517 nested

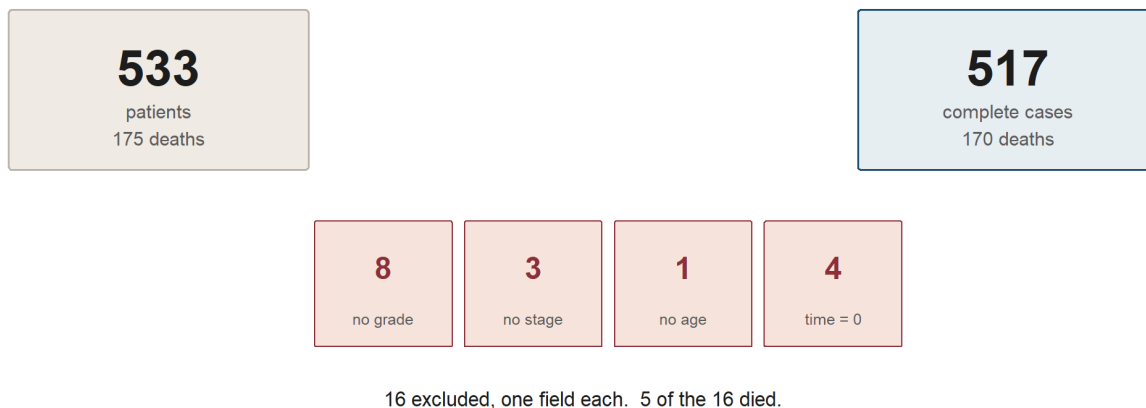


Figure 4: From 533 patients to the 517-patient nested cohort. The 16 exclusions are drawn separately: 8 lack grade, 3 lack stage, 1 lacks age, and 4 have survival time zero. 5 of those 16 died. Stage-only and grade-only sensitivities restore the grade-missing and stage-missing patients (525 and 520).

The original 541 tumor aliquots represented 533 patients; four patients contributed three aliquots each, accounting for the eight extra aliquots. No patient had more than one normal. Patient-level selection left 72 normals. Dropping the four multi-aliquot patients left 19 high-confidence genes, the first-barcode rule left 20, and summing tumor counts left 19. GRAMD1A, IFFO1, LTB4R, and RBM47 were absent from all three alternate sets. The deepest-library rule stays the primary rule. The unpaired GSE53757 gate kept 23

of the 23 primary genes and had Jaccard overlap 0.998 with the paired reproducible list. The paired DE fit included 72 patients and found 8,626 genes meeting the stated DE threshold. All 23 candidates retained their tumor-normal direction, while only 17 met the paired DE threshold; this is a sensitivity to the tumor-normal design, not an independent cohort. The primary analysis found 8,534 TCGA-significant mapped genes, 3,323 reproducible DEGs, 1,186 main-model FDR associations, 1,117 sensitivity-pass genes, 538 strict candidates, and 23 high-confidence candidates. Relative to the previous 27-gene run, 3 genes entered and 7 left the shortlist. The gene-level columns for all 23 candidates are in Table 1. Every candidate has the same columns.

Table 1: Supplementary Table 1. Evidence matrix for the 23 high-confidence candidates. Every row uses the same columns. Bold marks the GSE29609 reversals DDC and TCIRG1. An asterisk marks a gene lost under every alternate aliquot rule. Direction is relative to the TCGA hazard. Nested folds count how often the gene was the selected gene. TRACERx is the fraction of fixed-patient region draws that kept the TCGA direction.

Gene	TCGA log ₂ FC	TCGA HR	GSE40435	GSE53757	GSE29609	E-MTAB-1980	Marker	Purity	Folds	TRACERx	Aliquot
CYFIP2	-2.28	0.64	yes	yes	opposite	absent	yes	yes	22	94%	yes
CRYL1	-1.07	0.63	yes	yes	opposite	same	yes	yes	10	100%	yes
HHLA2	3.12	0.63	yes	yes	opposite	same	yes	yes	11	100%	yes
DDC	-1.70	0.66	yes	yes	opposite	same	no	yes	0	96%	yes
KL	-1.64	0.66	yes	yes	opposite	same	no	yes	1	37%	yes
RBM47*	-1.10	0.67	yes	yes	opposite	same	yes	yes	0	100%	no
PANK1	-1.49	0.62	yes	yes	opposite	same	yes	yes	0	84%	yes
GJB1	-1.47	0.65	yes	yes	same	same	no	yes	0	97%	yes
ACADM	-1.54	0.64	yes	yes	opposite	same	yes	yes	0	100%	yes
CLCN5	-1.43	0.66	yes	yes	opposite	same	no	yes	0	60%	yes
TMEM140	1.39	0.67	yes	yes	opposite	same	yes	yes	0	100%	yes
C1QTNF6	2.33	1.55	yes	yes	opposite	same	yes	yes	0	93%	yes
CADPS2	-1.63	0.66	yes	yes	same	same	yes	yes	0	100%	no
KNTC1	1.16	1.51	yes	yes	opposite	same	yes	yes	0	100%	yes
PODXL	-1.37	0.66	yes	yes	opposite	same	no	yes	0	100%	yes
HIBCH	-1.29	0.66	yes	yes	opposite	same	no	yes	0	99%	yes
TNIP1	1.05	0.66	yes	yes	opposite	same	yes	yes	0	69%	yes
TCIRG1	1.46	1.53	yes	yes	opposite	same	yes	yes	0	99%	yes
LTB4R*	2.59	1.51	yes	yes	same	same	yes	yes	0	39%	no
LRBA	-1.25	0.65	yes	yes	opposite	same	yes	yes	0	100%	yes
GRAMD1A*	1.32	1.50	yes	yes	same	same	yes	yes	0	98%	no
IFFO1*	1.78	1.50	yes	yes	absent	same	yes	yes	0	57%	no
TNFAIP2	1.29	1.53	yes	yes	same	opposite	yes	yes	0	64%	no

Figure 1 separates selection gates from the downstream assessments. The following benchmark uses the same 517 patients and splits.

The all-gene sensitivity modeled 32,192 count-QC genes and found 12,630 survival associations at global FDR below 0.05, illustrating how broad the outcome signal becomes without the upstream gate.

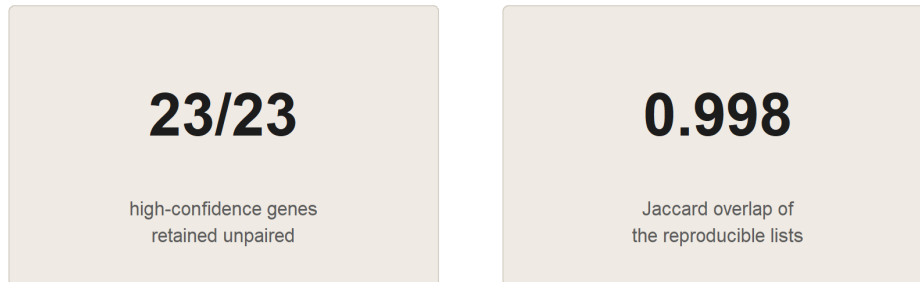
All 23 conditional bootstrap intervals excluded zero. The per-gene conditional CV showed positive mean concordance increments for 23 candidates, but both analyses start from genes chosen on the full TCGA outcome data. They cannot estimate the optimism of the complete selection process. The within-reproducible-gene overlap permutation did not show enrichment of the GEO-effect gate among survival-pass genes (empirical $p = 0.178$). It shuffles pass labels rather than refitting survival models, so it is a limited negative control.

B.2 Selection-aware prediction and funnel comparison

The nested analysis selected no gene in 0 of 50 training folds, but selected 9 different genes overall; the most frequent appeared in 22 folds. Thus, universal selection did not imply a stable gene identity.

2 folds required a no-replacement retry after a DESeq2 outlier-replacement error. Mean out-of-fold concordance was 0.744 for the clinical model and 0.748 after adding the training-selected gene, an increment of 0.0047 (patient-bootstrap 95% interval -0.0079 to 0.0174). The mean gene-minus-clinical three-year Brier-score difference was -0.0010. Mean risk-score calibration slopes were 0.86 and 0.76, respectively; the latter was farther from one despite the slightly lower average Brier score. The project prediction criterion failed. The clinical-only null selected a gene in 3 of 200 single-fold simulations. The permutation null, which keeps the training expression fits and scrambles training survival, selected a gene in 0 of 200 simulations. Neither frequency is a p-value for the ten-repeat mean increment.

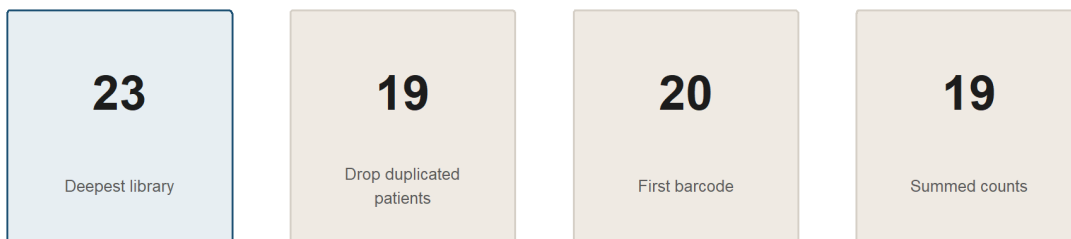
GSE53757



Paired versus unpaired. Adjacent rows do not prove patient identity.

Figure 5: Paired GSE53757 analysis versus the unpaired sensitivity. The unpaired gate retained 23 of 23 high-confidence genes. Jaccard overlap of the reproducible lists was 0.998. Adjacent-row pairing does not prove patient identity.

Aliquot rule



Lost under every alternate rule: GRAMD1A, IFFO1, LTB4R, RBM47

Figure 6: High-confidence genes under the primary deepest-library rule and three alternate aliquot rules. GRAMD1A, IFFO1, LTB4R, and RBM47 were lost under every alternate rule.

Genes selected inside the nested folds

Fifty training folds. The acceptance test adds only the top gene.

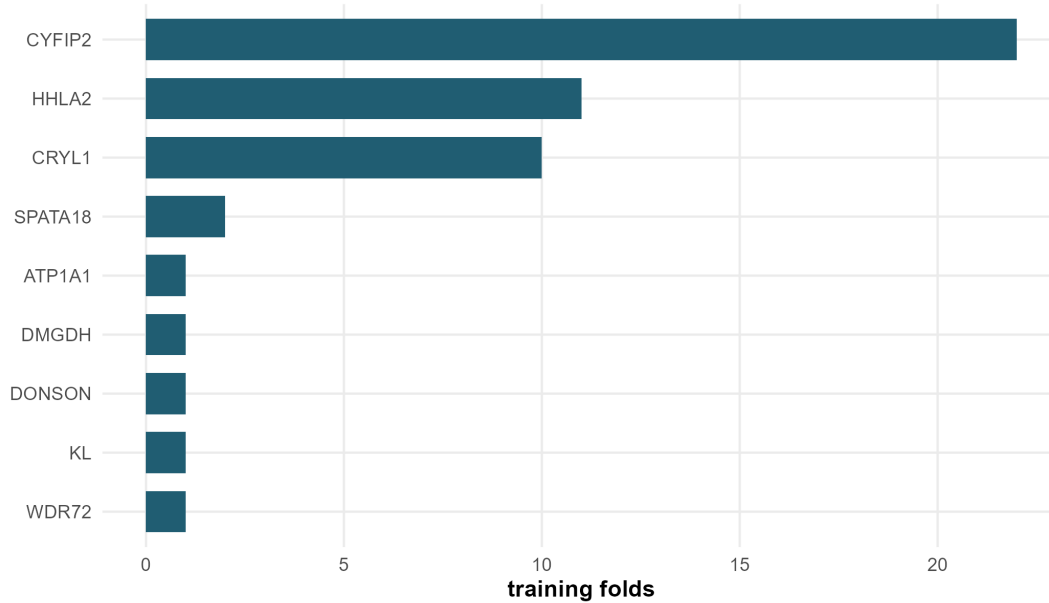


Figure 7: Training-fold selection frequency. 9 genes were chosen at least once. CYFIP2 was the top gene in 22 of 50 folds.

Selection-aware held-out increment

Dashed line: prespecified 0.01 increment threshold

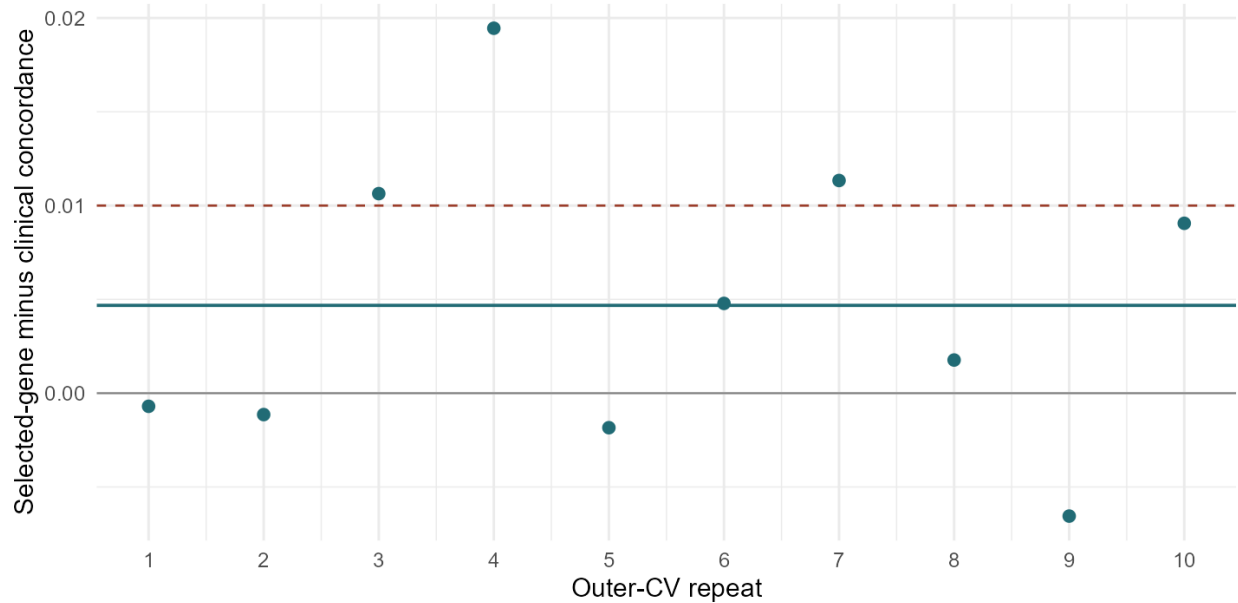


Figure 8: Concordance increment across ten complete outer-CV repeats. The dashed line marks the project 0.01 threshold; the patient-bootstrap interval is reported in text.

On those splits, the benchmark HYDRA arm matched the primary selected gene in every fold and again changed concordance by 0.0047 (interval -0.0069 to 0.0191). Survival-only selection, which used 20 genes, changed concordance by -0.0010 (interval -0.0150 to 0.0139). Differential-expression-only selection concentrated on 5 genes and changed concordance by -0.0014 (interval -0.0119 to 0.0079). Both left the mean three-year Brier score worse than the clinical model. The expression-matched control, spread across 44 genes, changed concordance by 0.0018 (interval 0.0001 to 0.0039). The interval sits above zero and remains well below 0.01, so it does not meet the project’s size requirement. Ridge regression kept the training-fold reproducible genes, on average 3,340 genes (range 3,057 to 3,447). Concordance rose by 0.0257 (interval 0.0115 to 0.0412), clearing the 0.01 line. The mean Brier difference was -0.0062 (interval -0.0141 to 0.0017), so the probability-score improvement is not stable under patient resampling. Calibration slopes were 0.86 for the clinical model, 0.76 after the single HYDRA gene, and 0.87 after the ridge model. The joint ridge model has a larger concordance increment than the one-gene point estimates. The increment compares the joint ridge predictor with a separately fit clinical model. It does not partial out the change in the clinical coefficients, and the penalty does not identify which genes carry the gain. Size factors, the log expression, the nonzero training-SD filter, and λ were all taken from the outer training fold. The ridge arm is not a validated signature. Because the comparison was specified after the one-gene rule, it does not replace the failed one-gene criterion. It was not evaluated on GSE29609 or E-MTAB-1980, so the concordance gain is a within-TCGA ranking result. ClearCode34, using 33 mapped genes, changed concordance by 0.0129 (interval -0.0018 to 0.0287) and the three-year Brier score by -0.0046. The calibration slope was 0.83. ClearCode34 is a published comparator, not a rerun of the one-gene acceptance test. Its interval includes zero and values below 0.01, so it does not establish a robust gain of at least 0.01. Table 2 puts every arm in one place. Table 3 records the ridge specification.

Table 2: Held-out benchmark on the same 517 patients and splits. Changes are versus a separately fit clinical model. The one-gene acceptance test is the HYDRA row only. Ridge is a description of ranking information in the reproducible set. ClearCode34 is a published comparator. It was not scored against the one-gene acceptance rule.

Arm	ΔC	95% interval	Δ Brier	Slope
HYDRA	+0.0047	-0.0069 to +0.0191	-0.0010	0.76
Survival-only	-0.0010	-0.0150 to +0.0139	+0.0029	0.71
DE-only	-0.0014	-0.0119 to +0.0079	+0.0025	0.81
Matched control	+0.0018	+0.0001 to +0.0039	-0.0006	0.86
ClearCode34	+0.0129	-0.0018 to +0.0287	-0.0046	0.83
Ridge	+0.0257	+0.0115 to +0.0412	-0.0062	0.87

Table 3: Supplementary Table 2. Ridge specification. This fit is not the one-gene acceptance test. The penalty does not select a gene list.

Item	Definition
Predictor set	training-fold reproducible DEGs with nonzero training standard deviation; penalty factor 1; none are removed by the penalty
L2 penalty	0
Inner CV	five event-stratified folds of the training patients only
Lambda selection	lambda.min
Tuning metric	partial-likelihood deviance inside the training fold; not the C-index and not the one-standard-error rule
Training-only preprocessing	outer-training DESeq2 size factors, $\log_2(\text{count} / \text{size factor} + 1)$, and the nonzero training-SD gene filter are fixed before <code>cv.glmnet</code>
Unpenalized clinical covariates	age, sex, stage, and grade; penalty factor 0; factor levels from the complete-case cohort; values from the training rows only

On E-MTAB-1980, adding the study *ccA/ccB* label to the limited clinical model changed in-sample

concordance by 0.0351 (interval -0.0483 to 0.1153). Adding CRYL1 changed it by 0.0034, and the signed panel of 22 mapped genes changed it by 0.0314. Those intervals include zero. The cohort has 23 deaths and was previously inspected.

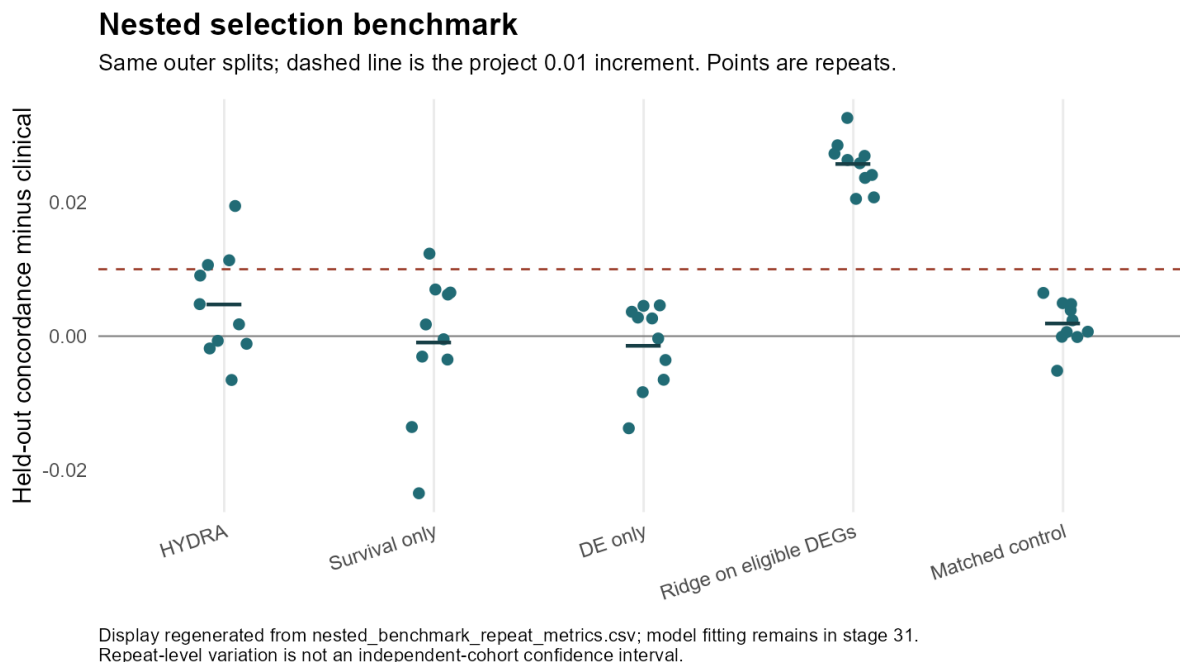


Figure 9: Repeat-level concordance increments for the five selectors fit inside the nested benchmark. The ClearCode34 comparison is reported in Table 2. The dashed line is the project 0.01 threshold.

The matched-list funnel criterion failed. In GSE29609 the complete-minus-survival-only directional-rate difference was 0.042 (paired gene-bootstrap 95% interval -0.194 to 0.270). The complete and survival-only lists mapped 20 and 19 genes there, respectively, so the observed difference compares proportions with distinct denominators. In E-MTAB-1980 both lists had 100% direction agreement among mapped genes (21 and 18, respectively), leaving no directional separation. Matching at 22 selected genes was necessary because one leave-one-GEO-out rule yielded only 22 eligible genes. The comparison reuses already inspected external outcomes and is not confirmatory.

B.3 External disagreement and composition

The harmonized comparison leaves 6 E-MTAB associations supported with age and grade, and 1 after adding T category. Neither adjusted GSE29609 model has same-direction FDR support or FDR-supported reversals. DDC and TCIRG1 nevertheless retain opposite effect directions; their age–grade–T heterogeneity tests have BH FDR below 0.05. These are exploratory differences in standardized cohort coefficients, not proof of a biological interaction. All 23 TCGA associations retain direction and FDR support across the completion scenarios, and 23 retain support after source-site stratification. Both plate models retain direction and FDR support for all 23 genes. CRYL1, RBM47, KNTC1, and TNFAIP2 fall below the absolute $\log_2$ fold-change cutoff of one in both models. The raw estimates, uncertainty, diagnostics, and scenarios are in the candidate ledger and dedicated CSVs.

ACADM, CRYL1, and DDC are three evidence profiles, not a ranked set of winners. ACADM has cohort-specific support in E-MTAB-1980: downregulated in TCGA and both GEO cohorts, with a protective TCGA hazard ratio. CRYL1 has a stronger main-model FDR than ACADM and is the high-confidence gene scored on E-MTAB-1980, because the nested procedure’s most frequent gene, CYFIP2, did not map to that array. DDC has a large GEO downregulation and a protective TCGA association, then reverses in GSE29609 at

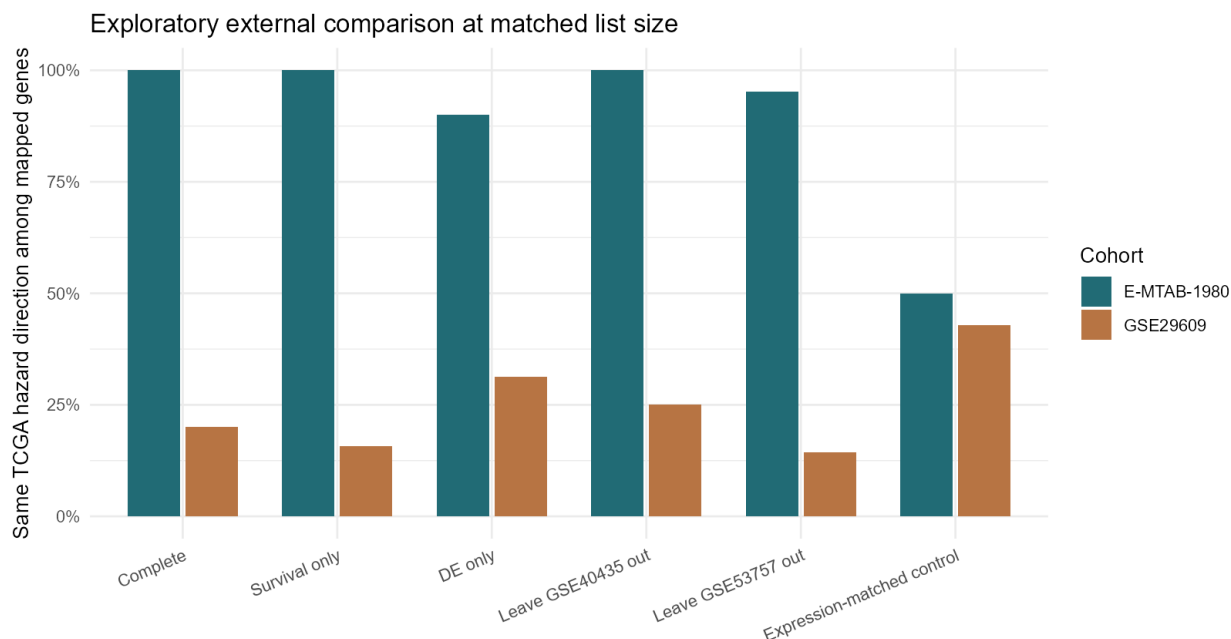


Figure 10: Same-direction agreement among mapped genes for six equal-sized lists in two previously inspected cohorts. Different platform coverage and event counts limit comparison. This comparison is not confirmatory.

Table 4: Exploratory harmonized adjustment of the fixed shortlist. Each cohort/model has a 23-hypothesis BH family, including unmapped genes. Same and opposite count FDR-supported effects relative to the primary TCGA direction; N counts patients. PH flags are nominal expression diagnostics, not exclusion gates. Grade and T are collapsed to low/high categories. These post-selection comparisons do not establish transportability.

Cohort	Model	Mapped	N	Events	Same	Opposite	PH
E-MTAB-1980	age + grade	22	99	23	6	0	4
E-MTAB-1980	age + grade + T	22	99	23	1	0	5
E-MTAB-1980	unadjusted	22	101	23	13	0	7
GSE29609	age + grade	21	39	17	0	0	2
GSE29609	age + grade + T	21	39	17	0	0	3
GSE29609	unadjusted	21	39	17	0	2	2
TCGA-KIRC	age + grade	23	520	171	23	0	0
TCGA-KIRC	age + grade + T	23	520	171	23	0	1
TCGA-KIRC	unadjusted	23	529	173	23	0	0

External survival does not copy TCGA

Shaded rows are DDC and TCIRG1. GSE29609 has 17 deaths; E-MTAB-1980 has 23.

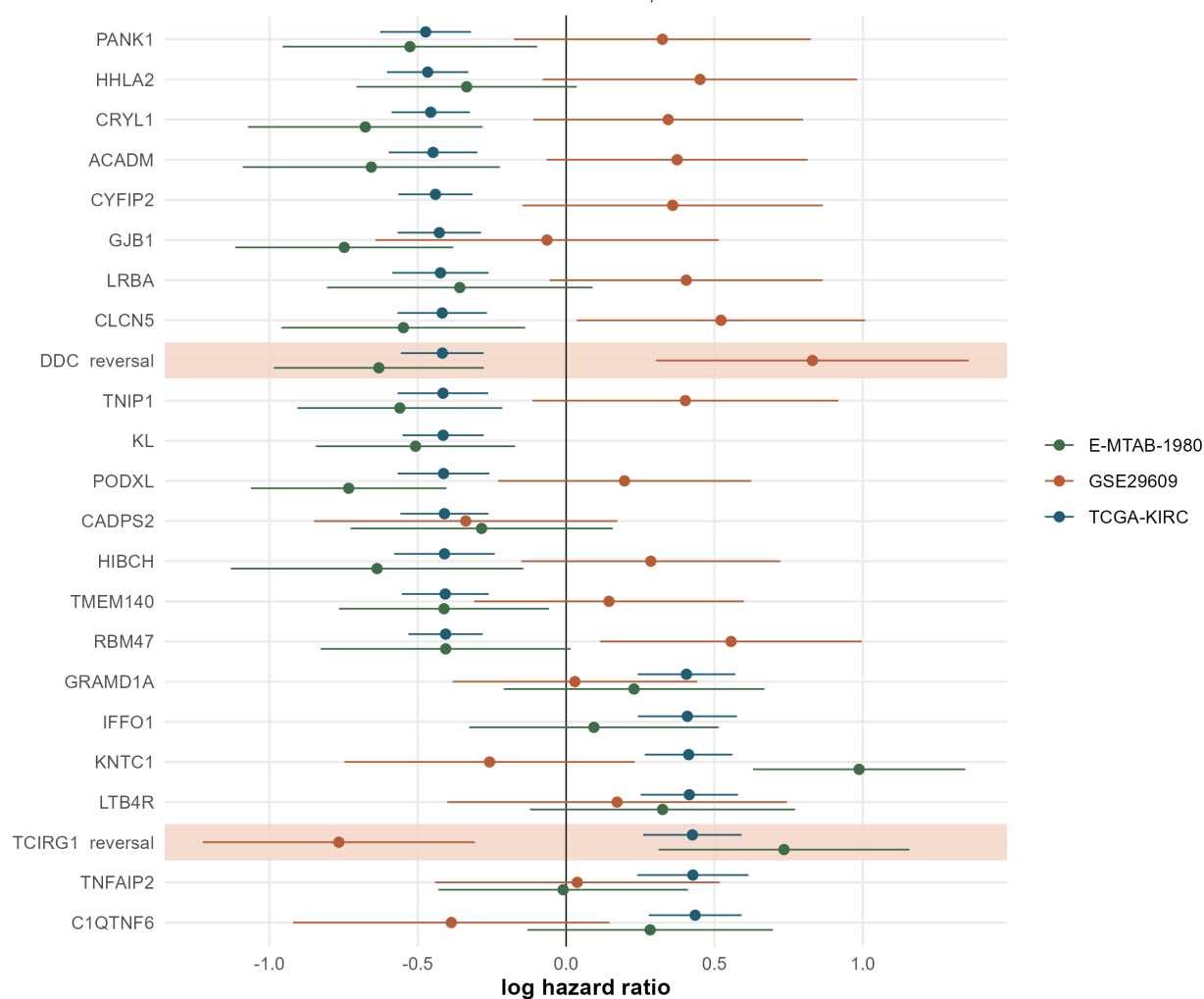


Figure 11: Log hazard ratios for the high-confidence genes in TCGA-KIRC, GSE29609 (17 deaths), and E-MTAB-1980 (23 deaths). Shaded rows are the DDC and TCIRG1 reversals. A shared direction is not an interval that excludes zero.

FDR below 0.05. The matrix records the same columns for the other genes. These are candidate associations, not mechanisms or clinical biomarkers.

GSE29609 contained 39 patients and 17 deaths. It mapped 21 primary candidates, of which 5 retained the TCGA hazard direction; 4 had nominal opposite-direction associations. DDC and TCIRG1 were reversed at FDR below 0.05. E-MTAB-1980 contained 101 patients and 23 deaths, mapped 22 candidates, retained the TCGA direction for 21, and gave 12 strict same-direction unadjusted and limited-adjustment FDR results. These cohorts do not support a claim of uniform external replication. GSE29609 has 17 deaths.

External direction

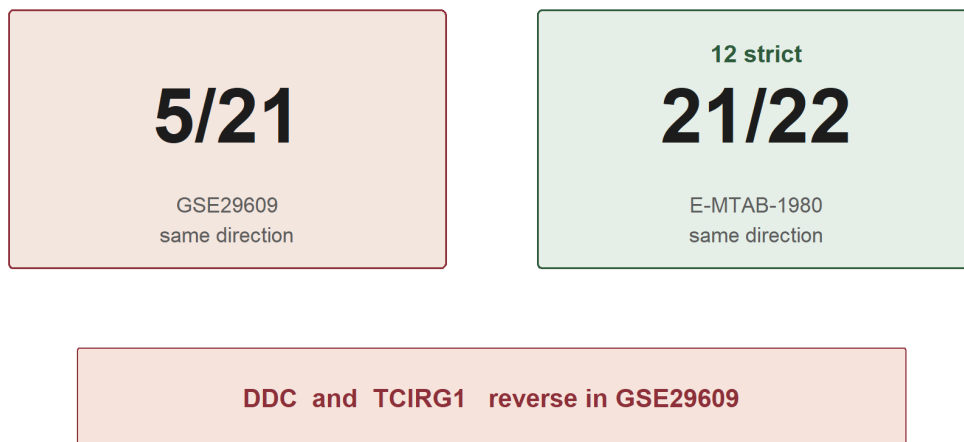


Figure 12: Same-direction counts among mapped high-confidence genes. GSE29609 retained the TCGA direction for 5 of 21. E-MTAB-1980 retained it for 21 of 22, with strict support for 12. DDC and TCIRG1 reverse in GSE29609.

Matched-patient marker-score adjustment removed FDR support for 6 candidates: CLCN5, DDC, GJB1, HIBCH, KL, PODXL. Direct consensus-purity adjustment retained direction and FDR support for all 23. The two adjustments do not agree, so composition control is model-dependent. Consensus purity remains an inferred bulk estimate rather than a tumor-cell measurement.

After correction for 23 tests, 0 nonlinear-expression comparisons and 0 expression-by-time comparisons had FDR below 0.05. Normal-tissue HPA profiles do not identify ccRCC tumor-cell origin.

B.4 Spatial sampling and randomized treatment

TRACERx linked 186 primary regions from 73 patients, including 16 deaths. A fixed event-stratified subset of 39 patients matches the size of GSE29609 and isolates region choice. Across genes, agreement was 98.5% [76.6–100.0] when those patients were fixed and only the chosen region varied, versus 86.3% [77.8–93.1] when both patient membership and region varied. Each smaller subset had 9 deaths. That event count makes this a sampling-variation check. It cannot attribute GSE29609’s reversals to spatial sampling alone.

The CheckMate 025 RNA subset included 250 patients (120 nivolumab and 130 everolimus); all 23 candidates were assessed. No expression-by-treatment interaction survived FDR correction for overall or progression-free survival in either the randomized or limited-adjustment model. Nominal interactions remain exploratory and do not establish treatment selection.

Composition adjustment of the 23 candidates

Triangles lose FDR support after marker-score adjustment. The line is 0.05.

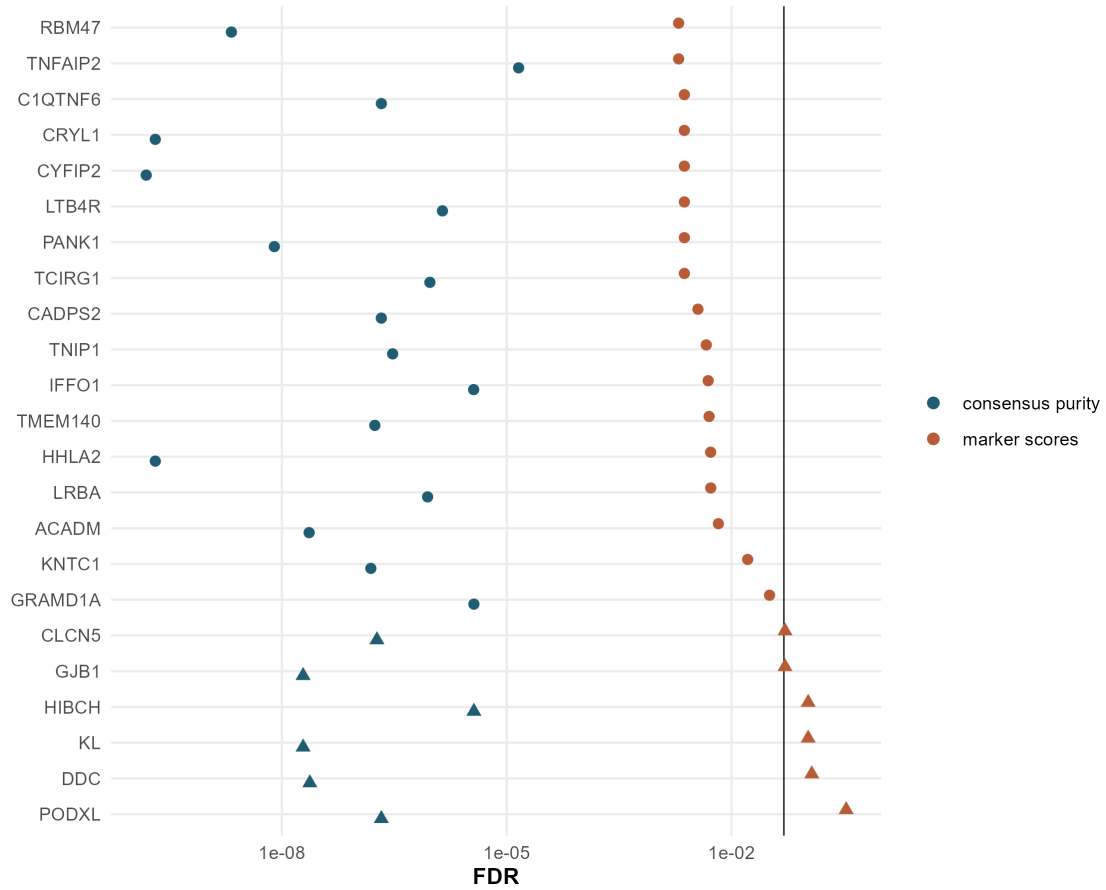


Figure 13: Marker-score adjustment and consensus-purity adjustment, shown together. Triangles mark the six genes that lose FDR support under the marker scores: CLCN5, DDC, GJB1, HIBCH, KL, PODXL. Consensus purity retains all 23. The vertical line is 0.05.

TRACERx direction

Panel labels are the median and interquartile range across genes.

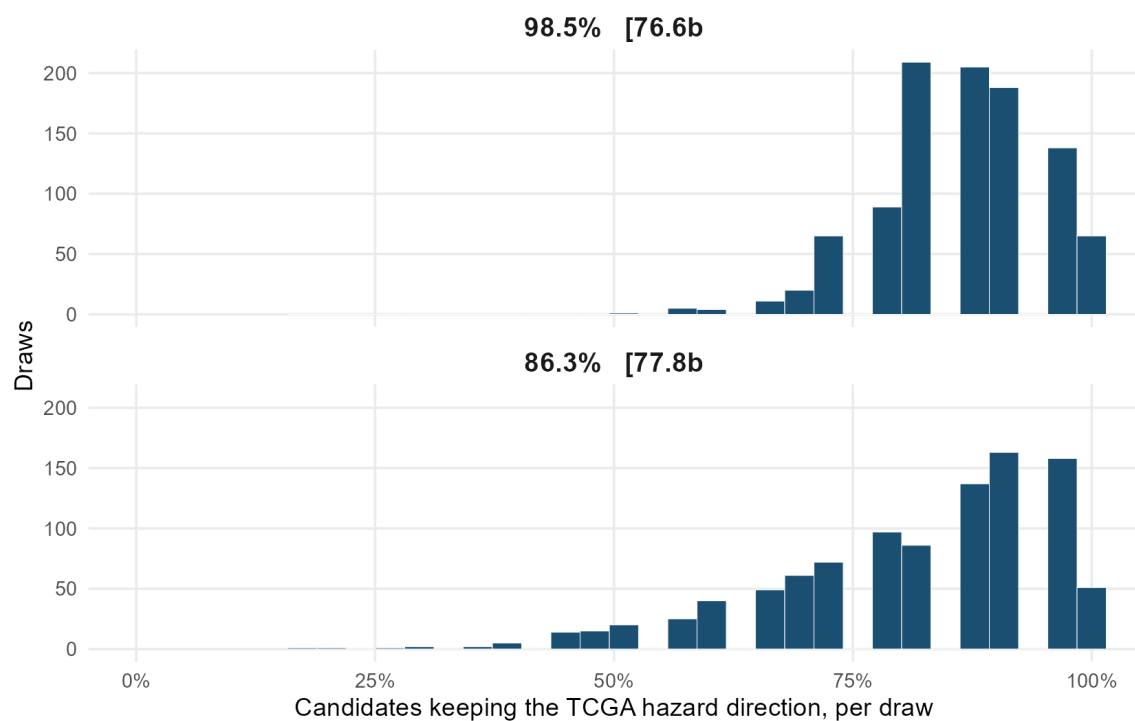


Figure 14: Fraction of high-confidence candidates that keep the TCGA hazard direction in 1,000 TRACERx draws. Across genes, agreement was 98.5% [76.6–100.0] with the 39 patients fixed, and 86.3% [77.8–93.1] when patient membership was also redrawn. Both subsets have 9 deaths. The brackets are interquartile ranges across genes.

C Complete priority association table

Table 5: Complete discovery shortlist. TCGA hazard ratios and 95% intervals are adjusted for age, sex, stage and grade. External hazard ratios are unadjusted. The marker column reports the composition-adjusted TCGA FDR; PH is the primary TCGA proportional-hazards diagnostic p-value. NA denotes unavailable evidence. None of these columns establishes clinical utility.

Gene	TCGA HR [95% CI]	TCGA FDR	E-MTAB HR	E-MTAB FDR	GSE29609 HR	GSE29609 FDR	Marker FDR	PH <i>p</i>
ACADM	0.64 [0.55, 0.74]	9.4e-07	0.52	0.0075	1.45	0.29	0.0067	0.24
C1QTNF6	1.55 [1.32, 1.81]	4.4e-06	1.33	0.22	0.68	0.3	0.0023	0.13
CADPS2	0.66 [0.57, 0.77]	4.9e-06	0.75	0.24	0.71	0.3	0.0036	0.48
CLCN5	0.66 [0.57, 0.77]	4.7e-06	0.58	0.019	1.69	0.18	0.052	0.82
CRYL1	0.63 [0.56, 0.72]	2e-08	0.51	0.0029	1.41	0.3	0.0023	0.18
CYFIP2	0.64 [0.57, 0.73]	1.6e-08	NA	NA	1.43	0.3	0.0023	0.11
DDC	0.66 [0.57, 0.76]	9.4e-07	0.53	0.0026	2.29	0.021	0.12	0.16
GJB1	0.65 [0.57, 0.75]	7.5e-07	0.47	0.00049	0.94	0.89	0.052	0.057
GRAMD1A	1.50 [1.27, 1.77]	3.8e-05	1.26	0.34	1.03	0.89	0.032	0.55
HHLA2	0.63 [0.55, 0.72]	3.1e-08	0.72	0.11	1.57	0.29	0.0053	0.056
HIBCH	0.66 [0.56, 0.79]	5.3e-05	0.53	0.023	1.33	0.3	0.11	0.31
IFFO1	1.50 [1.27, 1.78]	4.3e-05	1.10	0.69	NA	NA	0.0049	0.28
KL	0.66 [0.58, 0.76]	7.8e-07	0.60	0.0075	NA	NA	0.11	0.19
KNTC1	1.51 [1.30, 1.75]	4.1e-06	2.68	1.4e-06	0.77	0.42	0.016	0.65
LRBA	0.65 [0.56, 0.77]	1.3e-05	0.70	0.16	1.50	0.29	0.0053	0.37
LTB4R	1.51 [1.29, 1.78]	2.4e-05	1.38	0.2	1.19	0.65	0.0023	0.029
PANK1	0.62 [0.53, 0.73]	4.5e-07	0.59	0.03	1.38	0.3	0.0023	0.3
PODXL	0.66 [0.57, 0.77]	8.8e-06	0.48	0.00015	1.22	0.48	0.34	0.4
RBM47	0.67 [0.59, 0.76]	1.2e-07	0.67	0.093	1.74	0.094	0.002	0.04
TCIRG1	1.53 [1.30, 1.81]	1.9e-05	2.08	0.0029	0.46	0.021	0.0023	0.46
TMEM140	0.67 [0.57, 0.77]	4.4e-06	0.66	0.039	1.16	0.65	0.005	0.54
TNFAIP2	1.53 [1.27, 1.85]	0.00013	0.99	0.96	1.04	0.89	0.002	0.75
TNIP1	0.66 [0.57, 0.77]	6.2e-06	0.57	0.0046	1.49	0.3	0.0046	0.2